# PG No. 40 $C_{4h}(c)$ $4/m$ [ tetragonal ]

* Wyckoff site: `2a`, site symmetry: `4..`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `4..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | [0, 0, $Z$] | [0, 0, $z$] | `[1,2,3,4]` |
| 2 | [0, 0, $-Z$] | [0, 0, $-z$] | `[5,6,7,8]` |

Table 2: Wyckoff bond: `4b@2a`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | [$X$, $Y$, 0] | [0, 0, $z$] | `[1,-2]` |
| 2 | [$-Y$, $X$, 0] | [0, 0, $z$] | `[3,-4]` |
| 3 | [$-X$, $-Y$, 0] | [0, 0, $-z$] | `[5,-6]` |
| 4 | [$Y$, $-X$, 0] | [0, 0, $-z$] | `[7,-8]` |

Table 3: Wyckoff bond: `8c@2a`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | [$X$, $Y$, $Z$] | [0, 0, $z$] | `[1]` |
| 2 | [$-X$, $-Y$, $Z$] | [0, 0, $z$] | `[2]` |
| 3 | [$-Y$, $X$, $Z$] | [0, 0, $z$] | `[3]` |
| 4 | [$Y$, $-X$, $Z$] | [0, 0, $z$] | `[4]` |
| 5 | [$-X$, $-Y$, $-Z$] | [0, 0, $-z$] | `[5]` |
| 6 | [$X$, $Y$, $-Z$] | [0, 0, $-z$] | `[6]` |
| 7 | [$Y$, $-X$, $-Z$] | [0, 0, $-z$] | `[7]` |
| 8 | [$-Y$, $X$, $-Z$] | [0, 0, $-z$] | `[8]` |

* Wyckoff site: `4b`, site symmetry: `m..`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `m..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | [$X$, $Y$, 0] | [$x$, $y$, 0] | `[1,6]` |
| 2 | [$-X$, $-Y$, 0] | [$-x$, $-y$, 0] | `[2,5]` |
| 3 | [$-Y$, $X$, 0] | [$-y$, $x$, 0] | `[3,8]` |
| 4 | [$Y$, $-X$, 0] | [$y$, $-x$, 0] | `[4,7]` |

Table 5: Wyckoff bond: `4b@4b`, bond symmetry: `m..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | [0, 0, $Z$] | [$x$, $y$, 0] | `[1,-6]` |
| 2 | [0, 0, $Z$] | [$-x$, $-y$, 0] | `[2,-5]` |
| 3 | [0, 0, $Z$] | [$-y$, $x$, 0] | `[3,-8]` |
| 4 | [0, 0, $Z$] | [$y$, $-x$, 0] | `[4,-7]` |

Table 6: Wyckoff bond: `8c@4b`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, y, 0]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[-x, -y, 0]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $[-y, x, 0]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $[y, -x, 0]$ | `[4]` |
| 5 | $[-X, -Y, -Z]$ | $[-x, -y, 0]$ | `[5]` |
| 6 | $[X, Y, -Z]$ | $[x, y, 0]$ | `[6]` |
| 7 | $[Y, -X, -Z]$ | $[y, -x, 0]$ | `[7]` |
| 8 | $[-Y, X, -Z]$ | $[-y, x, 0]$ | `[8]` |

* Wyckoff site: `8c`, site symmetry: `1`

Table 7: Wyckoff bond: `8a@8c`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, y, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[-x, -y, z]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $[-y, x, z]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $[y, -x, z]$ | `[4]` |
| 5 | $[-X, -Y, -Z]$ | $[-x, -y, -z]$ | `[5]` |
| 6 | $[X, Y, -Z]$ | $[x, y, -z]$ | `[6]` |
| 7 | $[Y, -X, -Z]$ | $[y, -x, -z]$ | `[7]` |
| 8 | $[-Y, X, -Z]$ | $[-y, x, -z]$ | `[8]` |